

Use of Http Connector in Appworks

Steps to perform NoAuth Http Connector web service:

1. Create a CWS project in which you want to use the HTTP connector.
2. In the CWS project, add a Runtime Reference to the HTTP connector:
 - a. Create a new folder **Runtime Reference** in the project.
 - b. In this folder, add a runtime reference to an application connector.
 - c. From the folder **OpenText HTTP Connector**, select **OpenText HTTP Connector**, and then click **Finish**.
3. Add **HttpConnector** configuration as follows:

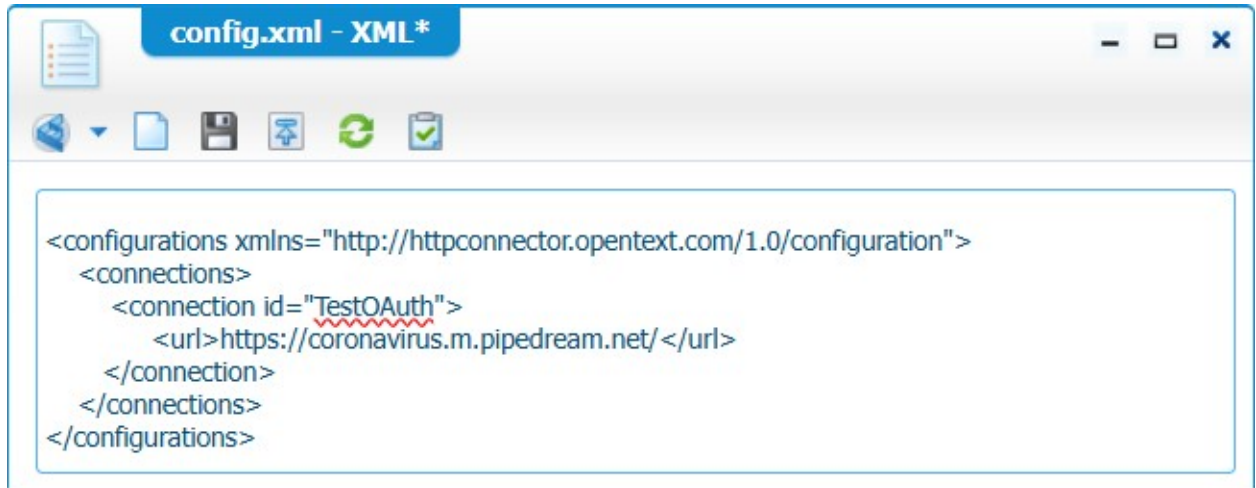
In runtime, the HTTP connector searches for configuration in the XML store under the **OpenText/HttpConnector** path. Hence, it is important to correctly define the XML store path in CWS.

 - a. Create a folder **XML Store**.
 - b. Right-click **XML Store**, and then select **Set Start Point of Qualified Name_Make**.
 - c. Add an XML Store Definition to the XML Store folder and name it **HttpConnector Config**.
 - d. Create a folder **OpenText** in the **XML Store** folder.
 - e. Open the **OpenText** folder and add a subfolder **HttpConnector**.
 - f. Open the **HttpConnector** folder and add the XML object **config.xml**.

This object contains the actual HTTP connector configuration.



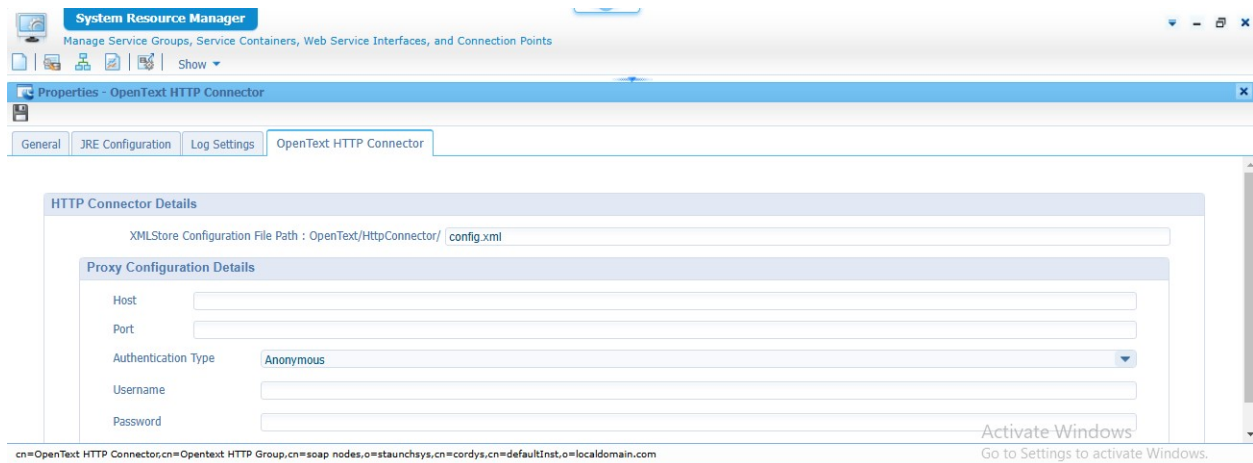
g. Add the following HTTP configuration and save the XML object.



Code :

```
<configurations
xmlns="http://httpconnector.opentext.com/1.0/configuration">
  <connections>
    <connection id="TestOAuth">
      <url>https://coronavirus.m.pipedream.net/</url>
    </connection>
  </connections>
</configurations>
```

Update the XML Store Configuration File Path in the System Resource Manager in the OpenText HTTP Connector Properties(Please refer to the mentioned screenshot).

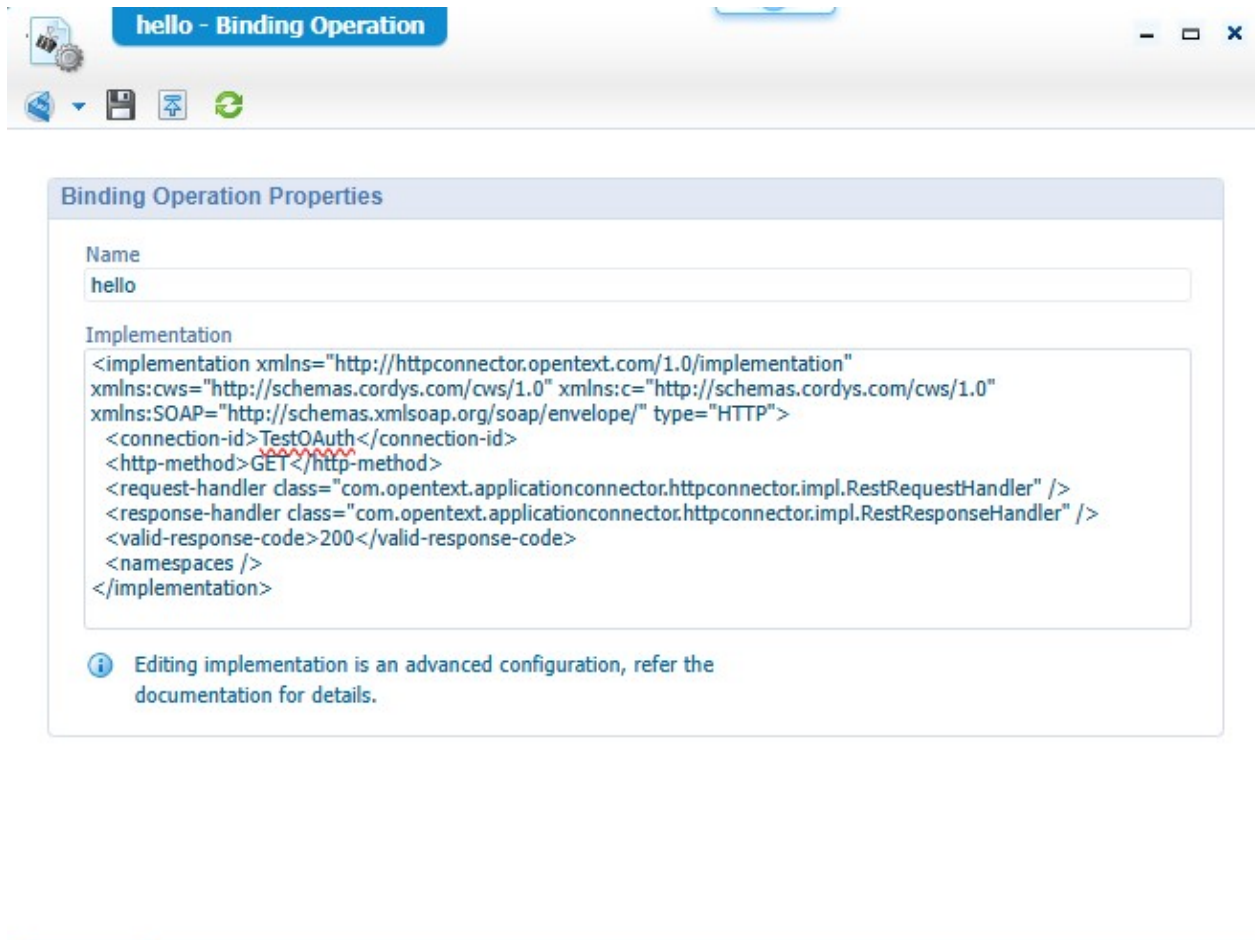


@id	Attribute of the connection tag and is the name of the connection.
url	Server endpoint URL. It can include a part or the complete URL of the service endpoint.
authenticate-always	Optional. Always sends the authentication information, also known as Preemptive Authentication. Default value is false.
check-certificate	Optional. Used with HTTPS connections only. It can be used as an option to check the SSL certificate. In a test environment, you can choose to ignore any errors while validating the SSL certificate. When it is set to false, any certificate that is either valid or invalid, expired, or not expired is accepted. Default value is false.
username	Optional. User name used for authentication at the service endpoint.
password	Optional. Password in Base64 encoding used for authentication at the service endpoint.
timeout	Optional. HTTP request timeout value in milliseconds. Default value is 30000 (30 seconds).

Add the Web service

1. From the project root, create a folder named **Web Services**.
2. Right-click the **Web Services** folder and add a Web service. Use **Custom Web Service** as the source.
3. Name the Web service with a name such as **GetEntries**.
4. Provide a **Description** or use the default description (same as the name).
5. Provide a **Namespace** for the service (Example: odes).
6. Type **GetEntries** as the **Web Service Interface Name**.
7. Select the **OpenText HTTP Connector** as the **Implementation Class**.
8. Add a Web service operation **GetEntries**.

9. Click **Finish**.
10. Open **GetEntries** and paste the following XML in the implementation field.



Code:

```
<implementation
xmlns="http://httpconnector.opentext.com/1.0/implementation"
xmlns:cws="http://schemas.cordys.com/cws/1.0"
xmlns:c="http://schemas.cordys.com/cws/1.0"
xmlns:SOAP="http://schemas.xmlsoap.org/soap/envelope/" type="HTTP">
```

```

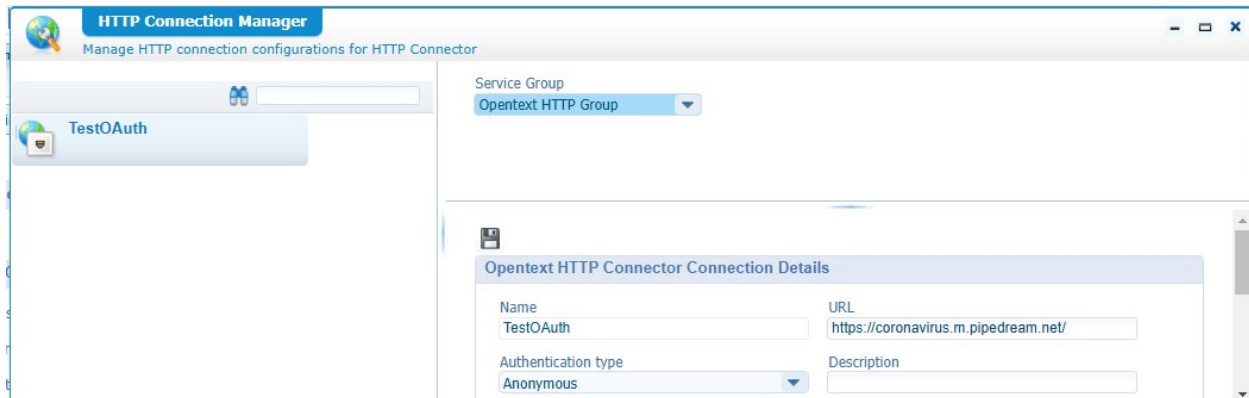
<connection-id>TestOAuth</connection-id>
<http-method>GET</http-method>
<request-handler
class="com.opentext.applicationconnector.httpconnector.impl.RestRequest
Handler" />
<response-handler
class="com.opentext.applicationconnector.httpconnector.impl.RestRespons
eHandler" />
<valid-response-code>200</valid-response-code>
<namespaces />
</implementation>

```

Note : Please use your Public API or use this link to do the same. <https://github.com/public-apis/public-apis?tab=readme-ov-file>

uri	Additional URL of the service endpoint. The service endpoint is a concatenation of the connection-id URL and the URI.
http-method	Used for GET method. HTTP connector also supports POST, PUT, DELETE, and PATCH.
request-handler class	The <code>RestRequestHandler</code> builds the correct REST request.
response-handler class	The <code>StandardResponseHandler</code> returns the response XML.
valid-response-code	Check if the HTTP response code is a valid one. This indicates whether the request was successful.
namespace	Parameter is empty and not used with the <code>RestRequestHandler</code> .

Check that the HTTP Connector Manager that your ID is connected or not:

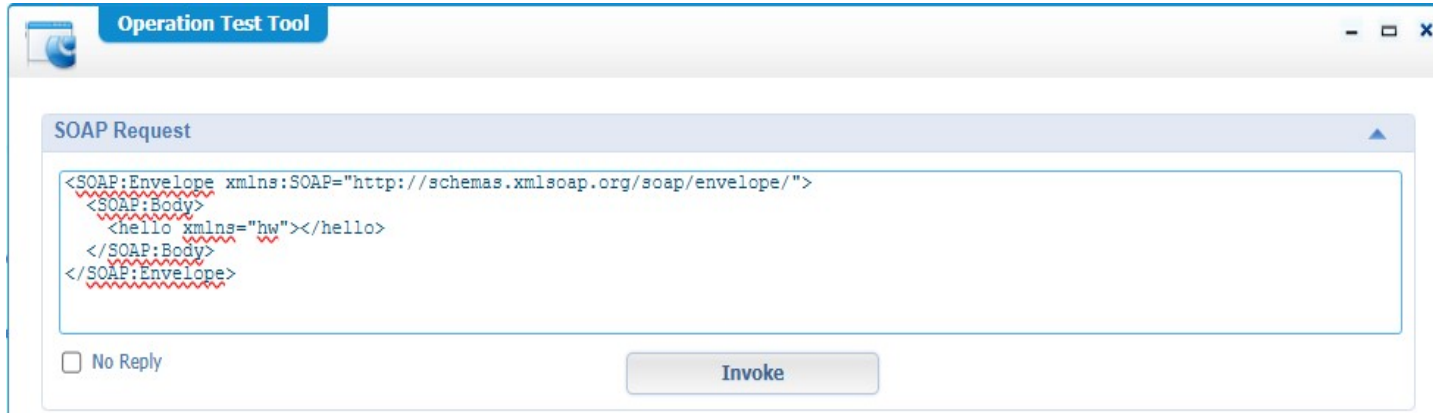


Please refer to the screenshot above

Run the Web service

1. Right-click the project root and select Publish to Organization. Part of this process is adding **config.xml** to the organization XML store.
2. In the System Resource Manager, create an HTTP Connector Service Group and service container if one does not already exist. Ensure that you select the interface you created when prompted and ensure that you add **config.xml** to the XMLStore Configuration File Path field.
3. Start the HTTP connector container to load the configuration. After changing and publishing a Web service implementation, use the Reset option to clear the internal method definition cache of the HTTP connector.
4. To test the Web service operation, right-click the GetEntriesWeb service operation, and then select Test Web Service Operation. The Operation Test Tool appears with a SOAP request for the GetEntriesWeb service operation.

5. Remove the placeholder PARAMETER text (this operation does not require additional input) and invoke the operation. The HTTP connector calls the GetEntriesRest service to get the requested entry info.



Code : **Sample SOAP request**

```
<SOAP:Envelope
xmlns:SOAP="http://schemas.xmlsoap.org/soap/envelope/">
  <SOAP:Body>
    <hello xmlns="hw"></hello>
  </SOAP:Body>
</SOAP:Envelope>
```

Code: **Sample SOAP response:**

```

<data>
  <helloResponse
xmlns:SOAP="http://schemas.xmlsoap.org/soap/envelope/"
xmlns="hw">
  <summaryStats>
    <global>
      <confirmed>676570149</confirmed>
      <recovered />
      <deaths>6881804</deaths>
    </global>
    <china>
      <confirmed>0</confirmed>
      <recovered>0</recovered>
      <deaths>0</deaths>
    </china>
    <nonChina>
      <confirmed>676570149</confirmed>
      <recovered />
      <deaths>6881804</deaths>
    </nonChina>
  </summaryStats>
  <cache>
    <lastUpdated>a minute ago</lastUpdated>
    <expires>in 4 minutes</expires>
    <lastUpdatedTimestamp>1731659348747</lastUpdatedTimestamp>
    <expiresTimestamp>1731659648747</expiresTimestamp>
  </cache>
  <dataSource>
    <url>https://raw.githubusercontent.com/CSSEGISandData/COVID-19/master/csse_covid_19_data/csse_covid_19_daily_reports/03-09-2023.csv</url>
    <lastGithubCommit>2023-03-10T04:51:58Z</lastGithubCommit>
    <publishedBy>John Hopkins University Center for Systems Science and Engineering</publishedBy>
    <ref>https://github.com/CSSEGISandData/COVID-19</ref>
  </dataSource>

  <apiSourceCode>https://pipedream.com/@/p_G6CLVM</apiSourceCode>
  >
    <rawData>
      <FIPS />
      <Admin2 />

```

```
<Province_State />
<Country_Region>Afghanistan</Country_Region>
<Last_Update>2023-03-10 04:21:03</Last_Update>
<Lat>33.93911</Lat>
<Long_>67.709953</Long_>
<Confirmed>209451</Confirmed>
<Deaths>7896</Deaths>
<Recovered />
<Active />
<Combined_Key>Afghanistan</Combined_Key>
<Incident_Rate>538.0424508714615</Incident_Rate>
<Case_Fatality_Ratio>3.76985547932452</Case_Fatality_Ratio>
</rawData>
<rawData>
  <FIPS />
  <Admin2 />
  <Province_State />
  <Country_Region>Zimbabwe</Country_Region>
  <Last_Update>2023-03-10 04:21:03</Last_Update>
  <Lat>-19.015438</Lat>
  <Long_>29.154857</Long_>
  <Confirmed>264276</Confirmed>
  <Deaths>5671</Deaths>
  <Recovered />
  <Active />
  <Combined_Key>Zimbabwe</Combined_Key>
  <Incident_Rate>1778.0885285919792</Incident_Rate>
  <Case_Fatality_Ratio>2.1458626587355645</Case_Fatality_Ratio>
</rawData>
</helloResponse>
</data>
```

Steps to perform OAuth Http Connector web service:

4. Create a CWS project in which you want to use the HTTP connector.
5. In the CWS project, add a Runtime Reference to the HTTP connector:
 - a. Create a new folder **Runtime Reference** in the project.
 - b. In this folder, add a runtime reference to an application connector.
 - c. From the folder **OpenText HTTP Connector**, select **OpenText HTTP Connector**, and then click **Finish**.
6. Add **HttpConnector** configuration as follows:

In runtime, the HTTP connector searches for configuration in the XML store under the **OpenText/HttpConnector** path. Hence, it is important to correctly define the XML store path in CWS.

 - a. Create a folder **XML Store**.
 - b. Right-click **XML Store**, and then select **Set Start Point of Qualified Name_Make**.
 - c. Add an XML Store Definition to the XML Store folder and name it **HttpConnector Config**.
 - d. Create a folder **OpenText** in the **XML Store** folder.
 - e. Open the **OpenText** folder and add a subfolder **HttpConnector**.
 - f. Open the **HttpConnector** folder and add the XML object **config.xml**.

This object contains the actual HTTP connector configuration.



- g. Add the following HTTP configuration and save the XML object.

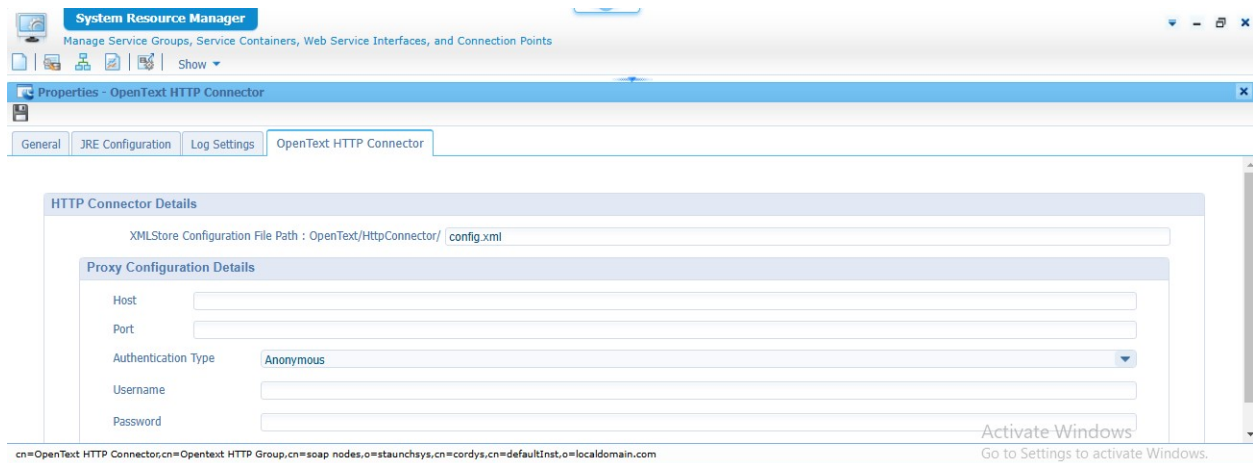


Code :

```
<?xml version="1.0" encoding="UTF-8"?>
<configurations
xmlns="http://httpconnector.opentext.com/1.0/configuration"
xmlns:SOAP="http://schemas.xmlsoap.org/soap/envelope/">
  <connections>
    <connection id="TestOAuth">
      <url>https://mkumar.onelogin.com/api/2/users</url>

    </connection>
  </connections>
</configurations>
```

Update the XML Store Configuration File Path in the System Resource Manager in the OpenText HTTP Connector Properties(Please refer to the mentioned screenshot).

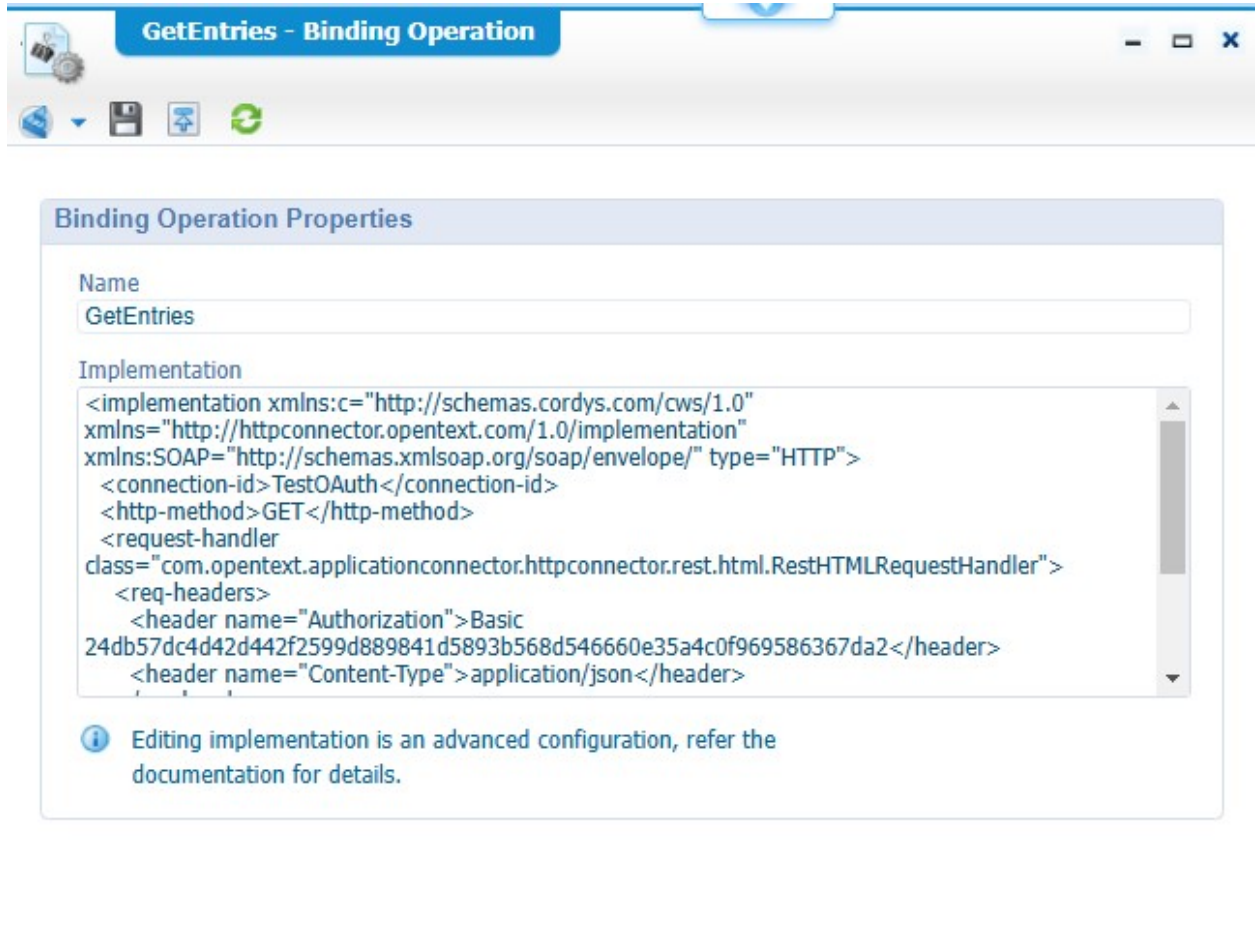


@id	Attribute of the connection tag and is the name of the connection.
url	Server endpoint URL. It can include a part or the complete URL of the service endpoint.
authenticate-always	Optional. Always sends the authentication information, also known as Preemptive Authentication. Default value is false.
check-certificate	Optional. Used with HTTPS connections only. It can be used as an option to check the SSL certificate. In a test environment, you can choose to ignore any errors while validating the SSL certificate. When it is set to false, any certificate that is either valid or invalid, expired, or not expired is accepted. Default value is false.
username	Optional. User name used for authentication at the service endpoint.
password	Optional. Password in Base64 encoding used for authentication at the service endpoint.
timeout	Optional. HTTP request timeout value in milliseconds. Default value is 30000 (30 seconds).

Add the Web service

- From the project root, create a folder named **Web Services**.
- Right-click the **Web Services** folder and add a Web service. Use **Custom Web Service** as the source.
- Name the Web service with a name such as **GetEntries**.
- Provide a **Description** or use the default description (same as the name).
- Provide a **Namespace** for the service (Example: odes).
- Type **GetEntries** as the **Web Service Interface Name**.
- Select the **OpenText HTTP Connector** as the **Implementation Class**.
- Add a Web service operation **GetEntries**.

19. Click **Finish**.
20. Open **GetEntries** and paste the following XML in the implementation field.



Code:

```
<implementation xmlns:c="http://schemas.cordys.com/cws/1.0"
xmlns="http://httpconnector.opentext.com/1.0/implementation"
xmlns:SOAP="http://schemas.xmlsoap.org/soap/envelope/" type="HTTP">
  <connection-id>TestOAuth</connection-id>
  <http-method>GET</http-method>
  <request-handler
class="com.opentext.applicationconnector.httpconnector.rest.html.RestHT
MLRequestHandler">
    <req-headers>
```

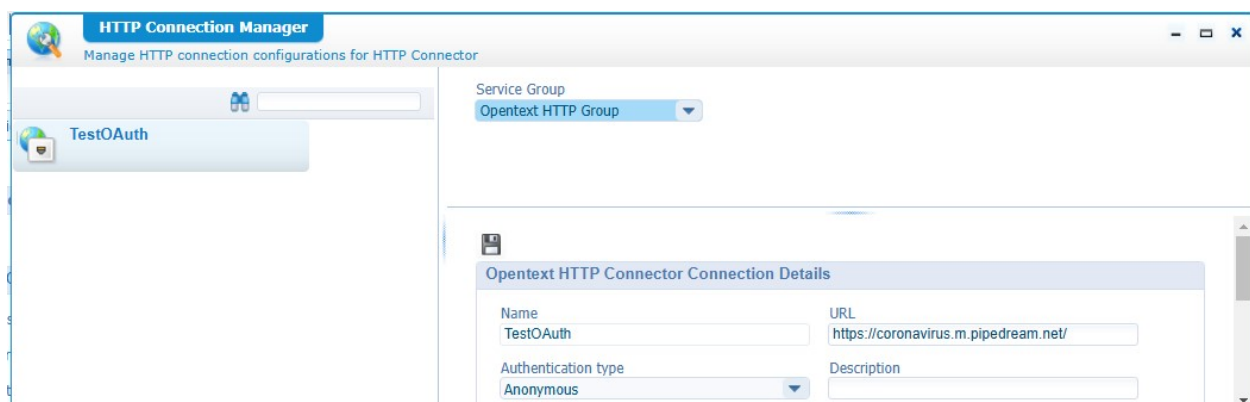
```

    <header name="Authorization">Basic
24db57dc4d42d442f2599d889841d5893b568d546660e35a4c0f969586367
da2</header>
    <header name="Content-Type">application/json</header>
</req-headers>
</request-handler>
<response-handler
class="com.opentext.applicationconnector.httpconnector.impl.RestRespons
eHandler" />
    <valid-response-code>200</valid-response-code>
<namespaces />
</implementation>

```

uri	Additional URL of the service endpoint. The service endpoint is a concatenation of the connection-id URL and the URI.
http-method	Used for GET method. HTTP connector also supports POST, PUT, DELETE, and PATCH.
request-handler class	The RestRequestHandler builds the correct REST request.
response-handler class	The StandardResponseHandler returns the response XML.
valid-response-code	Check if the HTTP response code is a valid one. This indicates whether the request was successful.
namespace	Parameter is empty and not used with the RestRequestHandler.

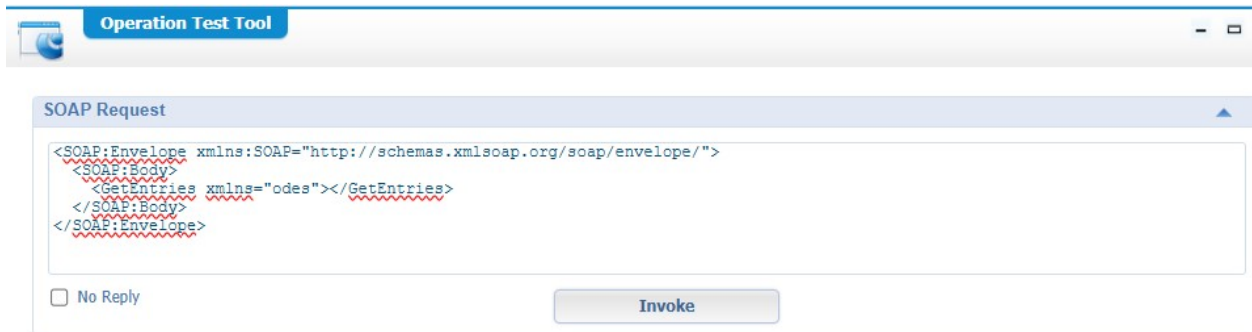
Check that the HTTP Connector Manager that your ID is connected or not:



Please refer to the screenshot above

Run the Web service

6. Right-click the project root and select Publish to Organization.
Part of this process is adding `config.xml` to the organization XML store.
7. In the System Resource Manager, create an HTTP Connector Service Group and service container if one does not already exist. Ensure that you select the interface you created when prompted and ensure that you add `config.xml` to the XMLStore Configuration File Path field.
8. Start the HTTP connector container to load the configuration. After changing and publishing a Web service implementation, use the Reset option to clear the internal method definition cache of the HTTP connector.
9. To test the Web service operation, right-click the GetEntries Web service operation, and then select Test Web Service Operation. The Operation Test Tool appears with a SOAP request for the GetEntries Web service operation.
10. Remove the placeholder PARAMETER text (this operation does not require additional input) and invoke the operation. The HTTP connector calls the GetEntries Rest service to get the requested entry info.



Code : Sample SOAP request

```
<SOAP:Envelope
xmlns:SOAP="http://schemas.xmlsoap.org/soap/envelope/">
  <SOAP:Body>
    <GetEntries xmlns="odes"></GetEntries>
  </SOAP:Body>
</SOAP:Envelope>
```

Code: Sample SOAP response:

```
<data>
  <GetEntriesResponse
xmlns:SOAP="http://schemas.xmlsoap.org/soap/envelope/"
xmlns="odes">
    <item>
      <invalid_login_attempts>0</invalid_login_attempts>
      <last_login>2024-11-18T04:41:51.996Z</last_login>
      <id>244085823</id>
      <lastname>Kumar</lastname>
      <status>1</status>
      <group_id />
      <distinguished_name />
      <member_of />
      <username />
      <locked_until />
      <activated_at>2024-11-15T11:20:20.372Z</activated_at>
      <email>mkumar@staunchsys.com</email>
      <created_at>2024-11-15T11:20:20.390Z</created_at>
```

```
<external_id />
<firstname>Manu</firstname>
<state>1</state>
<samaccountname />
<updated_at>2024-11-18T04:41:52.362Z</updated_at>
<invitation_sent_at>2024-11-
15T11:20:20.348Z</invitation_sent_at>
<directory_id />
<phone />
<password_changed_at>2024-11-
18T04:41:45.996Z</password_changed_at>
</item>
<item>
  <invalid_login_attempts>0</invalid_login_attempts>
  <last_login />
  <id>244086025</id>
  <lastname>qssad</lastname>
  <status>1</status>
  <group_id />
  <distinguished_name />
  <member_of />
  <username>sadsad</username>
  <locked_until />
  <activated_at>2024-11-15T11:47:08.587Z</activated_at>
  <email>sadsad@gmail.com</email>
  <created_at>2024-11-15T11:47:08.611Z</created_at>
  <external_id />
  <firstname>Pryan77</firstname>
  <state>1</state>
  <samaccountname />
  <updated_at>2024-11-15T11:47:08.611Z</updated_at>
  <invitation_sent_at />
  <directory_id />
  <phone>sdsadsadsa</phone>
  <password_changed_at />
</item>
</GetEntriesResponse>
</data>
```

